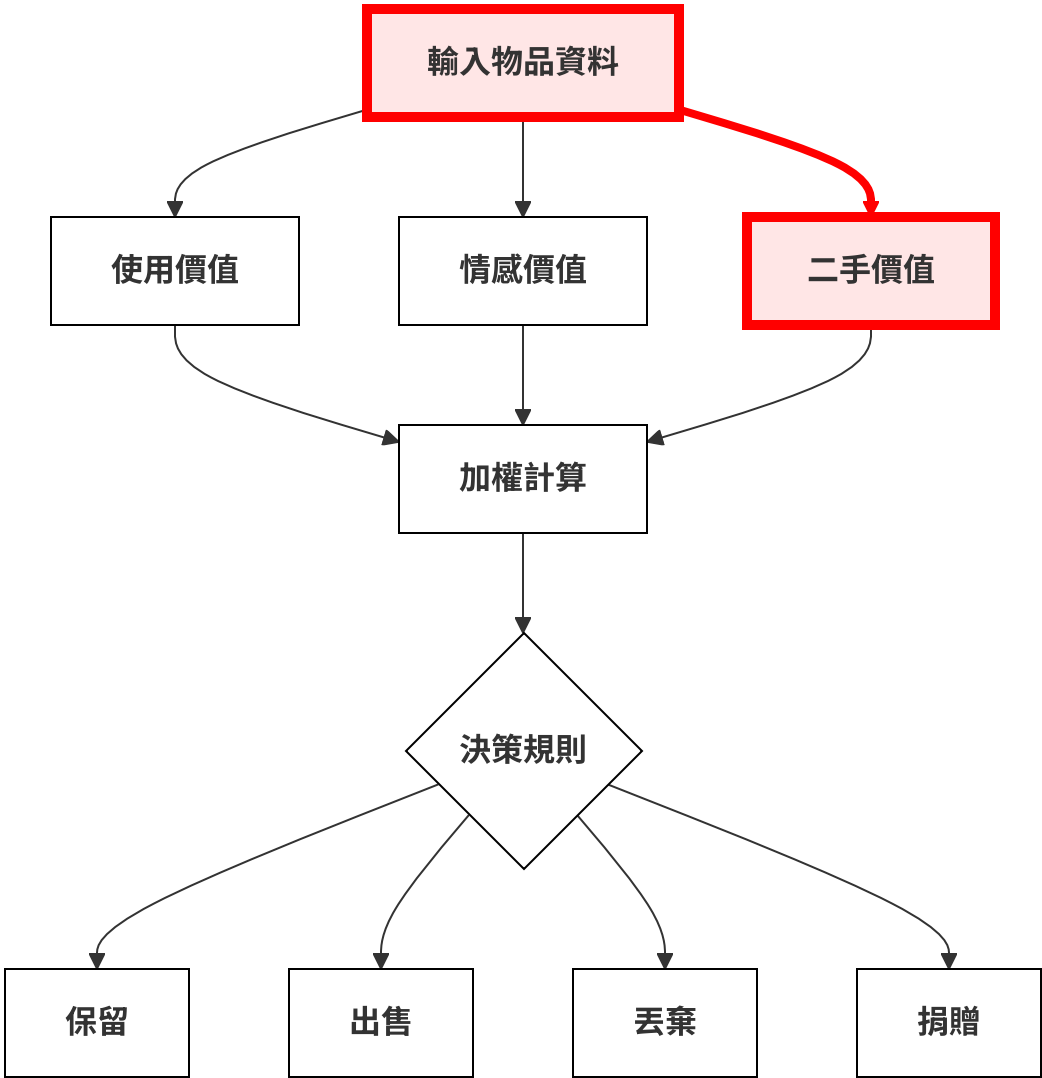


0. System Architecture



1. Problem & Dataset (Mercari Price Suggestion Challenge)

This project is based on the **Mercari Price Suggestion Challenge (Kaggle)** dataset.

Problem Definition

- Predict the final selling price of second-hand items
- Type: Regression problem
- Input: Product listing information
- Output: Price (USD)

Key Features

Each record represents a product listing:

- **name**: product title (e.g., "iPhone 11 64GB")
- **item_description**: detailed text description (NLP core feature)
- **category_name**: hierarchical category (e.g., Women / Tops)
- **brand_name**: brand (Nike, Apple, etc.)
- **item_condition_id**: condition level (new / used)
- **shipping**: shipping fee paid by seller or buyer
- **price (target)**: actual transaction price (label)

Dataset Characteristics

- ~1.5M training samples
- Highly noisy real-world e-commerce data
- Strong NLP + tabular feature interaction
- Missing values common (especially brand & category)
- Combines text + structured data + categorical features

2. Training and Model Saving

During training, the system automatically exports model weights and vectorizers to the `models_exported/` folder.

Run Training (by round)

- Round 1: `python main_ext.py 1`
- Round 2: `python main_ext.py 2`
- Round 3: `python main_ext.py 3`

After training, files such as `round_X_weights.pkl` and `round_X_vectorizer.pkl` will be generated. These files are used for prediction without retraining.

Merge Models

After completing rounds 1, 2, and 3, run:

```
python main_ext.py merge
```

3. Prediction Service

This project provides `predict_service.py`, which allows predictions using trained models without retraining.

How to Use

Activate your environment and run:

```
# Syntax
python predict_service.py --input [input_file] --output [output_file] --rounds [rounds]

# Example: predict test.tsv using round 1 model
python predict_service.py --input ../input/test.tsv --output results.csv --rounds 1
```

Parameters

- `--input` : **Required** input TSV file path
- `--output` : output file path (default: `predictions.csv`)
- `--rounds` : model rounds to use (e.g., `1` or `1,2,3`), default is `1`

4. Result

Validation Metrics Comparison

Model	RMSLE	R ²	MAE	RMSE
Round 1	0.3885	0.6188	8.6142	24.8363
Round 2	0.3908	0.6145	8.6451	24.9764
Round 3	0.3852	0.6287	8.5284	24.5101
Final Stacking (Lasso)	0.3757	0.6328	8.3075	24.3734

Prediction Results (Sample)

id	name	true_price	pred_1	pred_2	pred_3	final
0	MLB T Shirt	10.0	8.2175	8.7580	8.8224	8.4398
1	Razer Keyboard	52.0	65.5893	57.7846	52.6846	59.4399
2	AVA-VIV Blouse	10.0	8.9676	10.1834	10.2036	9.5192
3	Horse Statues	35.0	30.8875	30.9932	28.6670	30.6366
4	Gold Rose	44.0	23.7778	35.1438	31.8108	29.8407
5	Bundle Items	59.0	47.7234	69.4724	50.4827	55.3157
6	Acacia Top	64.0	62.1644	67.9720	55.9400	61.9939
7	Cheer Bundle	6.0	14.8646	10.8558	8.0851	10.6489
8	Nike Shorts	19.0	14.8536	17.0731	14.2864	15.0200
9	Clown Doll	8.0	11.4044	13.0452	10.6719	11.4454
10	Smashbox Primer	8.0	13.6681	13.4352	10.3010	12.4060